

SET THEORY AND ALGEBRA

(2 Marks)

1. GATE CSE 2025 Set 2

MCQ (More than One Correct Answer) Marks: +2 / -0

Let F be the set of all functions from $\{1, \dots, n\}$ to $\{0, 1\}$. Define the binary relation $\preceq$ on F as follows:

$\forall f, g \in F, f \preceq g$ if and only if $\forall x \in \{1, \dots, n\}, f(x) \leq g(x)$, where $0 = 1$.

Which of the following statement(s) is/are TRUE? re TRUE?

- (A) $\preceq$ is a symmetric relation
- (B) $(F, \preceq)$ is a partial order
- (C) $(F, \preceq)$ is a lattice
- (D) $\preceq$ is an equivalence relation

2. GATE CSE 2025 Set 1

MCQ (More than One Correct Answer) Marks: +2 / -0

$A = \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in F$ we define

$$(f_1 \odot f_2)(n) = f_1(n) + f_2(n)$$

for every number n in A . Which of the following is/are CORRECT about the mathematical structure $(F, \odot)$?

- (A) $(F, \odot)$ is an Abelian group.
- (B) $(F, \odot)$ is an Abelian monoid.
- (C) $(F, \odot)$ is a non-Abelian group.
- (D) $(F, \odot)$ is a non-Abelian monoid.

3. GATE CSE 2024 Set 2

Numerical Marks: +2 / -0

Let Z_n be the group of integers $\{0, 1, 2, \dots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is _____.

4. GATE CSE 2024 Set 1

MCQ (More than One Correct Answer) Marks: +2 / -0

Consider the operators $\diamond$ and $\square$ defined by $a \diamond b = a + 2b$, $a \square b = ab$, for positive integers. Which of the following statements is/are TRUE?

- (A) Operator $\diamond$ obeys the associative law
- (B) Operator $\square$ obeys the associative law
- (C) Operator $\diamond$ over the operator $\square$ obeys the distributive law
- (D) Operator $\square$ over the operator $\diamond$ obeys the distributive law

5. GATE CSE 2023

MCQ (More than One Correct Answer) Marks: +2 / -0

Let $f : A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation $\sim$ on the set A as

$$a_1 \sim a_2 \text{ if } f(a_1) = f(a_2),$$

where $a_1, a_2 \in A$. Let $\varepsilon = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sim$. Define a new mapping $F : \varepsilon \rightarrow B$ as

$$F([x]) = f(x), \text{ for all the equivalence classes } [x] \text{ in } \varepsilon.$$

Which of the following statements is/are TRUE?

- (A) F is NOT well-defined.
- (B) F is an onto (or surjective) function.
- (C) F is a one-to-one (or injective) function.
- (D) F is a bijective function.

6. GATE CSE 2023

MCQ (More than One Correct Answer) Marks: +2 / -0

Let X be a set and 2^X denote the powerset of X . Define a binary operation Δ on 2^X as follows:

$$A \Delta B = (A - B) \cup (B - A).$$

Let $H = (2^X, \Delta)$. Which of the following statements about H is/are correct?

- (A) H is a group.
- (B) Every element in H has an inverse, but H is NOT a group.
- (C) For every $A \in 2^X$, the inverse of A is the complement of A .
- (D) For every $A \in 2^X$, the inverse of A is A .

7. GATE CSE 2021 Set 2

Numerical Marks: +2 / -0

Let S be a set consisting of 10 elements. The number of tuples of the form (A, B) such that A and B are subsets of S , and $A \subseteq B$ is _____.

8. GATE CSE 2021 Set 1

MCQ (More than One Correct Answer) Marks: +2 / -0

A relation R is said to be circular if aRb and bRc together imply cRa . Which of the following options is/are correct?

- (A) If a relation S is transitive and circular, then S is an equivalence relation.
- (B) If a relation S is reflexive and symmetric, then S is an equivalence relation.
- (C) if a relation S is reflexive and circular, then S is an equivalence relation.
- (D) if a relation S is circular and symmetric, then S is an equivalence relation.

9. GATE CSE 2018

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let N be the set of natural numbers. Consider the following sets.

P : Set of Rational numbers (positive and negative)

Q : Set of functions from $\{0, 1\}$ to N

R : Set of functions from N to $\{0, 1\}$

S : Set of finite subsets of N .

Which of the sets above are countable?

- (A) Q and S only
- (B) P and S only
- (C) P and R only
- (D) P, Q and S only

10. GATE CSE 2016 Set 1

Numerical Marks: +2 / -0

A function $f : N^+ \rightarrow N^+$, defined on the set of positive integers N^+ , satisfies the following properties:

$$f(n) = f(n/2) \text{ if } n \text{ is even}$$

$$f(n) = f(n+5) \text{ if } n \text{ is odd}$$

Let $R = \{i \mid \exists j : f(j) = i\}$ be the set of distinct values that f takes. The maximum possible size of R is _____.

11. GATE CSE 2016 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements:

- I. Each compound in $U \setminus S$ reacts with an odd number of compounds.
- II. At least one compound in $U \setminus S$ reacts with an odd number of compounds.
- III. Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is **ALWAYS TRUE**?

- (A) Only I
- (B) Only II
- (C) Only III
- (D) None

12. GATE CSE 2015 Set 3

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let R be a relation on the set of ordered pairs of positive integers such that $((p, q), (r, s)) \in R$ if and only if $p - s = q - r$. Which one of the following is true about R ?

- (A) Both reflexive and symmetric
- (B) Reflexive but not symmetric
- (C) Not reflexive but symmetric
- (D) Neither reflexive nor symmetric

13. GATE CSE 2015 Set 2

Numerical Marks: +2 / -0

Let X and Y denote the sets containing 2 and 20 distinct objects respectively and F denote the set of all possible functions defined from X to Y . Let f be randomly chosen from F . The probability of f being one-to-one is _____.

14. GATE CSE 2015 Set 2

Numerical Marks: +2 / -0

The number of onto functions (surjective functions) from set $X = \{1, 2, 3, 4\}$ to set $Y = \{a, b, c\}$ is _____.

15. GATE CSE 2014 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the following relation on subsets of the set S integers between 1 and 2014. For two distinct subsets U and V of S we say $U < V$ if the minimum element in the symmetric difference of the two sets is in U .

Consider the following two statements:

S1 There is a subset of S that is larger than every other subset.

S2: There is a subset of S that is smaller than every other subset.

Which one of the following is **CORRECT**?

- (A) Both S1 and S2 are true
- (B) S1 is true and S2 is false
- (C) S2 is true and S1 is false
- (D) Neither S1 nor S2 is true

16. GATE CSE 2014 Set 1

Numerical Marks: +2 / -0

Let S denote the set of all functions $f : \{0, 1\}^4 \rightarrow \{0, 1\}$.
Denote by N the number of functions from S to the set $\{0, 1\}$.
The value of $\log_2 \log_2 N$ is _____

17. GATE CSE 2014 Set 3

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the set of all functions $f : \{0, 1, \dots, 2014\} \rightarrow \{0, 1, \dots, 2014\}$ such that $f(f(i)) = i$, for all $0 \leq i \leq 2014$.

Consider the following statements:

P . For each such function it must be the case that for every

$i, f(i) = i$

Q . For each such function it must be the case that for some

$i, f(i) = 1$

R . Each such function must be onto.

Which one of the following is CORRECT?

- (A) P, Q and R are true
- (B) Only Q and R are true
- (C) Only P and Q are true
- (D) Only R is true

18. GATE CSE 2014 Set 3

Numerical Marks: +2 / -0

There are two elements x, y in a group $(G, *)$ such that every element in the group can be written as a product of some number of x 's and y 's in some order. It is known that $x * x = y * y = x * y * x * y = y * x * y * x = e$ where e is the identity element. The maximum number of elements in such a group is _____.

19. GATE CSE 2012

MCQ (Single Correct Answer) Marks: +2 / -0.6

How many onto (or surjective) functions are there from an n -element ($n \geq 2$) set to a 2-element set?

- (A) 2^n
- (B) $2^n - 1$
- (C) $2^n - 2$
- (D) $2(2^n - 2)$

20. GATE CSE 2009

MCQ (Single Correct Answer) Marks: +2 / -0.6

For the composite table of a cyclic group shown below

	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	b	a
d	d	c	a	b

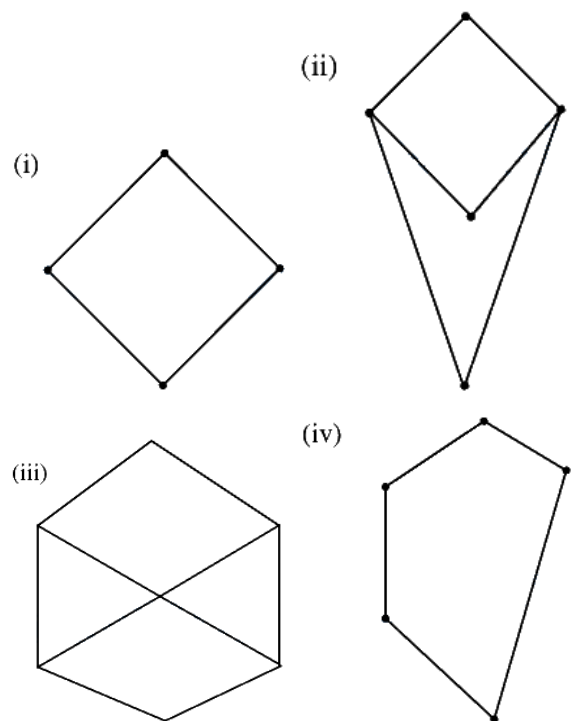
Which one of the following choices is correct?

- (A) a, b are generators
- (B) b, c are generators
- (C) c, d are generators
- (D) d, a are generators

21. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the following Hasse diagrams.



Which all of the above represent a lattice?

- (A) (i) and (iv) only
- (B) (ii) and (iii) only
- (C) (iii) only
- (D) (i), (ii) and (iv) only

22. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +2 / -0.6

A partial order P is defined on the set of natural numbers as following. Herw x/y denotes integer division.

- i) $(0, 0) \in P$.
 ii) $(a, b) \in P$ if and only $a \% 10 \leq b \% 10$ and $a/10, b/10 \in P$.

Consider the following ordered pairs:

- i) (101, 22) ii) (22, 101)
 iii) (145, 265) iv) (0, 153)

Which of these ordered pairs of natural numbers are contained in P ?

- (A) (i), (iii) and (iv)
 (B) (ii) and (iv)
 (C) (i) and (iv)
 (D) (iii) and (iv)

23. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the set of (column) vectors defined by $X = \{x \in R^3 \mid x_1 + x_2 + x_3 = 0, \text{ where } x^T = [x_1 + x_2 + x_3]^T\}$.

Which of the following is TRUE?

- (A) $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a basis for the subspace X .
 (B) $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X .
 (C) X is not a subspace of R^3 .
 (D) None of the above.

24. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +2 / -0.6

How many different non-isomorphic Abelian groups of order 4 are there?

- (A) 2
 (B) 3
 (C) 4
 (D) 5

25. GATE CSE 2006

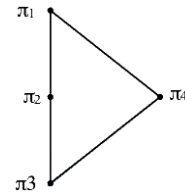
MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the set $S = \{a, b, c, d\}$. Consider the following 4 partitions $\pi_1, \pi_2, \pi_3, \pi_4$ on S : $\pi_1 = \{\overline{abcd}\}$, $\pi_2 = \{\overline{ab}, \overline{cd}\}$, $\pi_3 = \{\overline{abc}, \overline{d}\}$, $\pi_4 = \{\overline{a}, \overline{b}, \overline{c}, \overline{d}\}$. Let $\prec$ be the partial order on the set of partitions $S' = \{\pi_1, \pi_2, \pi_3, \pi_4\}$ defined as follows: $\pi_i \prec \pi_j$ if and only if π_i refines π_j . The poset diagram for $(S', \prec)$ is

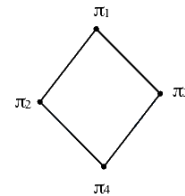
(A)



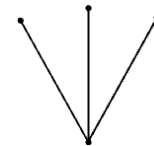
(B)



(C)



(D)

**26. GATE CSE 2006**

MCQ (Single Correct Answer) Marks: +2 / -0.6

Given a set of elements $N = \{1, 2, \dots, n\}$ and two arbitrary subsets $A \subseteq N$ and $B \subseteq N$, how many of the $n!$ permutations π from N to N satisfy $\min(\pi(A)) = \min(\pi(B))$, where $\min(S)$ is the smallest integer in the set of integers S , and $\pi(S)$ is the set of integers obtained by applying permutation π to each element of S ?

- (A) $(n - |A \cup B|) |A| |B|$
 (B) $(|A|^2 + |B|^2) n^2$
 (C) $n! \frac{|A \cap B|}{|A \cup B|}$
 (D) $\frac{|A \cap B|^2}{\binom{n}{|A \cup B|}}$

27. GATE CSE 2006

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let $S = \{1, 2, 3, \dots, m\}$, $m > 3$. Let $X_1, \dots, X_n$ be subsets of S each of size 3. Define a function f from S to the set of natural numbers as, $f(i)$ is the number of sets X_j that contain the element i . That is $f(i) = |\{j \mid i \in X_j\}|$.

Then $\sum_{i=1}^m f(i)$ is

- (A) $3m$
- (B) $3n$
- (C) $2m + 1$
- (D) $2n + 1$

28. GATE CSE 2006

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let E, F and G be finite sets.

Let $X = (E \cap F) - (F \cap G)$

and $Y = (E - (E \cap G)) - (E - F)$. Which one of the following is true?

- (A) $X \subset Y$
- (B) $X \supset Y$
- (C) $X = Y$
- (D) $X - Y \neq \emptyset$ and $X - Y \neq \emptyset$

29. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let A be a set with n elements. Let C be a collection of distinct subsets of A such that for any two subsets S_1 and S_2 in C , either $S_1 \subset S_2$ or $S_2 \subset S_1$. What is the maximum cardinality of C ?

- (A) n
- (B) $n + 1$
- (C) $2^{n-1} + 1$
- (D) $n!$

30. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let R and S be any two equivalence relations on a non-empty set A . Which one of the following statements is TRUE?

- (A) $R \cup S$, $R \cap S$ are both equivalence relations
- (B) $R \cup S$ is an equivalence relations
- (C) $R \cap S$ is an equivalence relations
- (D) Neither $R \cup S$ nor $R \cap S$ is an equivalence relation

31. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let $f: B \rightarrow C$ and $g: A \rightarrow B$ be two functions and let $h = f \circ g$. Given that h is an onto function which one of the following is TRUE?

- (A) f and g should both be onto functions
- (B) f should be onto but g need not be into
- (C) g should be onto but f need not be onto
- (D) both f and need not be onto

32. GATE CSE 2004

MCQ (Single Correct Answer) Marks: +2 / -0.6

The inclusion of which of the following sets into

$S = \{\{1, 2\}, \{1, 2, 3\}, \{1, 3, 5\}, \{1, 2, 4\}, \{1, 2, 3, 4, 5\}\}$

is necessary and sufficient to make S a complete lattice under the partial order defined by set containment?

- (A) $\{1\}$
- (B) $\{1\}, \{2, 3\}$
- (C) $\{1\}, \{1, 3\}$
- (D) $\{1\}, \{1, 3\}, \{1, 2, 3, 4\}, \{1, 2, 3, 5\}$

33. GATE CSE 2004

MCQ (Single Correct Answer) Marks: +2 / -0.6

The following is the incomplete operation table of a 4-element group.

*	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b				
c				

The last row of the table is

- (A) $c \ a \ e \ b$
- (B) $c \ b \ a \ e$
- (C) $c \ b \ e \ a$
- (D) $c \ e \ a \ b$

34. GATE CSE 2002

MCQ (Single Correct Answer) Marks: +2 / -0.6

The binary relation $S = \phi$ (empty set) on set $A = \{1, 2, 3\}$ is

- (A) Neither reflexive nor symmetric
- (B) Symmetric and reflexive
- (C) Transitive and reflexive
- (D) Transitive and symmetric

35. GATE CSE 2001

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let $f : A \rightarrow B$ be a function, and let E and F be subsets of A . Consider the following statements about images.

$$S1 : f(E \cup F) = f(E) \cup f(F)$$

$$S2 : f(E \cap F) = f(E) \cap f(F)$$

Which of the following is true about $S1$ and $S2$?

- (A) Only $S1$ is correct
- (B) Only $S2$ is correct
- (C) Both $S1$ and $S2$ are correct
- (D) None of $S1$ and $S2$ is correct

36. GATE CSE 2001

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the following statements:

$S1$: There exist infinite sets A, B, C such that

$A \cap (B \cup C)$ is finite.

$S2$: There exist two irrational numbers x and y such that $(x + y)$ is rational.

Which of the following is true about $S1$ and $S2$?

- (A) Only $S1$ is correct
- (B) Only $S2$ is correct
- (C) Both $S1$ and $S2$ are correct
- (D) None of $S1$ and $S2$ is correct

37. GATE CSE 2000

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let $P(S)$ denote the power set of a set S . Which of the following is always true?

- (A) $P(P(S)) = P(S)$
- (B) $P(S) \cap P(P(S)) = \{\emptyset\}$
- (C) $P(S) \cap S = P(S)$
- (D) $S \notin P(S)$

38. GATE CSE 2000

MCQ (Single Correct Answer) Marks: +2 / -0.6

A relation R is defined on the set of integers as xRy if $f(x + y)$ is even. Which of the following statements is true?

- (A) R is not an equivalence relation
- (B) R is an equivalence relation having 1 equivalence class
- (C) R is an equivalence relation having 2 equivalence classes
- (D) R is an equivalence relation having 3 equivalence classes

39. GATE CSE 1999

Subjective Marks: +2 / -0

(a) Mr. X claims the following:

If a relation R is both symmetric and transitive, then R is reflexive. For this, Mr. X offers the following proof.

"From xRy , using symmetry we get yRx . Now because R is transitive, xRy and yRx together imply xRx . Therefore, R is reflexive."

Briefly point out the flaw in Mr. X's proof.

(b) Give an example of a relation R which is symmetric and transitive but not reflexive.

40. GATE CSE 1998

MCQ (Single Correct Answer) Marks: +2 / -0.6

The binary relation $R = \{(1, 1), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4)\}$ on the set $A = \{1, 2, 3, 4\}$ is

- (A) Reflexive, symmetric and transitive
- (B) Neither reflexive, nor irreflexive but transitive
- (C) Irreflexive, symmetric and transitive
- (D) Irreflexive and antisymmetric

41. GATE CSE 1998

Subjective Marks: +2 / -0

Let $(A, *)$ be a semigroup. Furthermore, for every a and b in A , if $a \neq b$, then $a * b \neq b * a$.

(a) Show that for every a in A

$$a * a = a$$

(b) Show that for every a, b in A

$$a * b * a = a$$

(c) Show that for every a, b, c in A

$$a * b * c = a * c$$

42. GATE CSE 1998

Subjective Marks: +2 / -0

Suppose $A = \{a, b, c, d\}$ and Π_1 is the following partition of A

$$\Pi_1 = \{\{a, b, c\}, \{d\}\}$$

(a) List the ordered pairs of the equivalence relations induced by Π_1

(b) Draw the graph of the above equivalence relation.

43. GATE CSE 1996

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let R denote the set of real numbers. Let $f: R \times R \rightarrow R \times R$ be a bijective function defined by $f(x, y) = (x + y, x - y)$. The inverse function of f is given by

$$(A) f^{-1}(x, y) = \left(\frac{1}{x + y}, \frac{1}{x - y} \right)$$

$$(B) f^{-1}(x, y) = (x - y, x + y)$$

$$(C) f^{-1}(x, y) = \left(\frac{x + y}{2}, \frac{x - y}{2} \right)$$

$$(D) f^{-1}(x, y) = (2(x - y), 2(x + y))$$

44. GATE CSE 1996

MCQ (Single Correct Answer) Marks: +2 / -0.6

Which one of the following is false?

- (A) The set of all bijective functions on a finite set forms a group under function composition.
- (B) The set $\{1, 2, \dots, p-1\}$ forms a group under multiplication mod p where p is a prime number.
- (C) The set of all strings over a finite alphabet Σ forms a group under concatenation.
- (D) A subset $s \neq \phi$ of G is a subgroup of the group if and only if for any pair of elements $a, b \in s$, $a * b^{-1} \in s$.

45. GATE CSE 1996

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let R be a non-empty relation on a collection of sets defined by $A^R B$ if and only if $A \cap B = \phi$. Then, (pick the true statement)

- (A) R is reflexive and transitive
- (B) R is symmetric and not transitive
- (C) R is an equivalence relation
- (D) R is not reflexive and not symmetric

46. GATE CSE 1995

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let A be the set of all nonsingular matrices over real numbers and let $*$ be the matrix multiplication operator. Then

- (A) A is closed under $*$ but $\langle A, * \rangle$ is not a semigroup.
- (B) $\langle A, * \rangle$ is a semigroup but not a monoid.
- (C) $\langle A, * \rangle$ is a monoid but not a group.
- (D) $\langle A, * \rangle$ is a group but not an abelian group

47. GATE CSE 1994

MCQ (Single Correct Answer) Marks: +2 / -0.6

Some group (G, o) is known to be abelian. Then, which one of the following is true for G ?

- (A) $g = g^{-1}$ for every $g \in G$.
- (B) $g = g^2$ for every $g \in G$.
- (C) $(goh)^2 = g^2 o h^2$ for every $g, h \in G$.
- (D) G is of finite order.

48. GATE CSE 1989

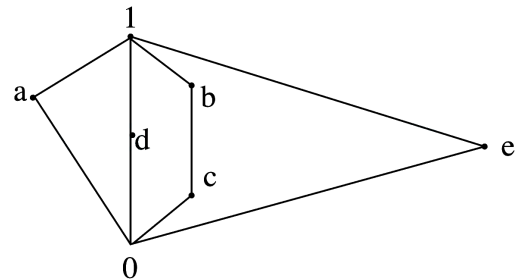
Fill in the Blanks Marks: +2 / -0

The transitive closure of the relation

 $\{(1, 2) (2, 3) (3, 4) (5, 4)\}$ on the set $A = \{1, 2, 3, 4, 5\}$ is _____ .**49. GATE CSE 1988**

Fill in the Blanks Marks: +2 / -0

The complement(s) of the element 'a' in the lattice shown in Fig. is (are)



SET THEORY AND ALGEBRA (5 Marks)

50. GATE CSE 2002

Subjective Marks: +5 / -0

Let A be a set of n (> 0) elements. Let N_r be the number of binary relations on A and Let N_f be the number of functions from A to A .

- (a) Give the expression for N_r in terms of n .
- (b) Give the expression for N_f in terms of n .
- (c) Which is larger for all possible n , N_r or N_f ?

51. GATE CSE 2002

Subjective Marks: +5 / -0

- (a) $S = \{< 1, 2 >, < 2, 1 >\}$ is binary relation on set $A = \{1, 2, 3\}$. Is it irreflexive?

Add the minimum number of ordered pairs to S to make it an equivalence relation. Give the modified S .

- (b) Let $S = \{a, b\}$ and let $\square S$ be the power set of S . Consider the binary relation ' $\subseteq$ (set inclusion)' on $\square S$. Draw the Hasse diagram corresponding to the lattice $(\square(S), \subseteq)$

52. GATE CSE 2000

Subjective Marks: +5 / -0

A multiset is an unordered collection of elements where elements may repeat any number of times. The size of a multiset is the number of elements in it counting repetitions.

- (a) what is the number of multisets of size 4 that can be constructed from n distinct elements so that at least one element occurs exactly twice?
- (b) How many multisets can be constructed from n distinct elements?

53. GATE CSE 2000

Subjective Marks: +5 / -0

Let $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and $\otimes$ denote multiplication modulo 8, that is, $x \otimes y = (xy) \bmod 8$

- (a) Prove that $(0, 1, \otimes)$ is not a group.
- (b) Write 3 distinct groups $(G, \otimes)$ where $G \subset S$ and G has 2 elements.

54. GATE CSE 1995

Subjective Marks: +5 / -0

Let G_1 and G_2 be subgroups of a group G .

- (a) Show that $G_1 \cap G_2$ is also a subgroup of G .
- (b) Is $G_1 \cup G_2$ always a subgroup of G ?

55. GATE CSE 1992

Subjective Marks: +5 / -0

- (a) If G is a group of even order, then show that there exists an element $a \neq e$, the identifier e , such that $a^2 = e$

(b) Consider the set of integers $\{1, 2, 3, 4, 6, 8, 12, 24\}$ together with the two binary operations LCM (lowest common multiple) and GCD (greatest common divisor). Which of the following algebraic structures does this represent?

- i) Group ii) ring
 - iii) field iv) lattice
- Justify your answer
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